

SkillBridge Data Architecture & Ingestion Specifications

Reference Document: Web Scrapers, Public REST APIs, Skills Telemetry & Lifecycle Management (Zero-Key Configuration)

This document details all live data ingestion methods, scrapers, APIs, and dynamic calculation engines active in the SkillBridge platform. It excludes third-party services requiring paid or custom API keys (e.g. Adzuna, Joooble) and documents how live job opportunities, skill demand scores, and district-level employment telemetry are continuously sourced without hardcoded datasets.

1. Web Scrapers (Government & Public Recruitment)

Four automated scrapers harvest real-time Indian central, state, and Maharashtra-specific public sector recruitment postings:

#	Target Source & URL	Scraper Technique & Library	Extracted Fields & Data Scope
1	IndGovtJobs https://www.indgovtjobs.in/feeds/posts/default?alt=rss	XML / RSS Scraper Python requests + xml.etree.ElementTree	Scrapes active recruitment drives for MAHA Metro Rail Corporation, PSU Engineering, Bank IT Officers, and Central/State Ministries . Extracts job titles, role descriptions, publication dates, and primary application URLs.
2	FreeJobAlert https://www.freejobalert.com/feed/	XML / RSS Scraper Python requests + xml.etree.ElementTree	Scrapes public recruitments across IITs, AIIMS, IOCL, State Technical Boards, and PSUs . Normalizes location fields and categorizes engineering/technical posts.
3	National Career Service (NCS) https://www.ncs.gov.in/NCSPages/VacancyList.aspx	HTML DOM Scraper Python requests + BeautifulSoup4	Targets Ministry of Labour & Employment portal vacancy DOM nodes (div.job-item, tr.vacancy-row). Extracts verified vacancy titles, hiring organizations, Maharashtra state tags, and official NCS apply links.
4	Mahaswayam Portal https://rojgar.mahaswayam.gov.in/MarketplaceJobList	HTML DOM Scraper Python requests + BeautifulSoup4	Scrapes Maharashtra State Skill Development & Employment Department marketplace listings. Parses card elements, employer names, local district tags, and direct application URLs.

2. Public REST APIs (Zero API Keys Required)

Open JSON endpoints providing global, Indian, and remote technology listings:

#	Provider & Endpoint	Protocol & Format	Ingestion Details
5	RemoteOK https://remoteok.com/api	Public REST / JSON No Auth Token	Ingests 80–100+ active software engineering, full-stack, AI, DevOps, data science, and cloud jobs. Ingests company profile, salary ranges (where declared), tech tags, and application links.
6	Jobicy https://jobicy.com/api/v2/remote-jobs?count=100	Public REST / JSON No Auth Token	Pulls up to 100 developer, frontend/backend, QA, and product positions. Extracts job role, seniority level, tech stack tags, and application links open to Indian candidates.

7	Remotive https://remotive.com/api/remote-jobs	Public REST / JSON No Auth Token	Multi-category remote tech listings. Extracts job title, company name, requirements, skills array, and direct application redirection links.
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3. Market Demand Intelligence & Skills Telemetry

- Provider:** GitHub Search API (Public REST)
- Endpoint:** https://api.github.com/search/repositories?q={skill}+stars:>50
- Methodology:** Rather than using static survey data, the backend queries real-time developer activity across **24 tracked technical skills** (Python, React, AWS, Docker, Kubernetes, AI/ML, SQL, etc.).
- **Demand Score:** Logarithmically scaled (0–100) based on repository creation velocity and star traction.
 - **Dynamic Trend Badges:** Automatically classifies each skill into HOT (≥ 85), RISING (≥ 70), STABLE (≥ 50), or DECLINING (< 50).

4. Maharashtra District Intelligence Engine

- Calculation Paradigm:** Fully on-the-fly aggregation without static hardcoded figures.
- Monitored Districts:** Pune, Mumbai, Nagpur, Nashik, Aurangabad, Solapur, Kolhapur, Amravati, Thane, Nanded.
- Metrics Computed Real-Time:**
1. **Primary Hiring Sector:** Evaluated dynamically by clustering active job postings in the district into IT & Software, Manufacturing, Automobile, Healthcare, BFSI, or Logistics.
 2. **Demand Index:** Computed from active job posting volume and hiring frequency.
 3. **Top Demanded Skill:** Identified via keyword extraction of skill tags across local job vacancies.
 4. **Training Capacity & Status:** Compares active job openings to baseline institutional capacity to output status tags (CRITICAL_SHORTAGE, HIGH_DEMAND, BALANCED) and recommended training seat quotas.

5. Expiry Cleanup & Deduplication Lifecycle

- **Deduplication:** Fingerprinted by normalized (title + company_name). Updates existing records rather than inserting redundant rows on repeated synchronizations.
- **Active Tracking:** Every ingestion cycle updates the last_seen_at timestamp for active postings.
- **Dynamic Purge Logic:** Expired postings are deactivated when unobserved across feeds for > 14 days, preventing dead application links while guarding against transient network interruptions.
- **Re-hiring Auto-Restoration:** When an expired role reappears on official portals, the record is immediately re-activated.